

GRADUATE CERTIFICATE IN PATTERN RECOGNITION SYSTEMS

PRACTICE MODULE

Stackable Graduate Certificate Programme in Artificial Intelligence

Intelligent Reasoning Systems	Pattern Recognition Systems	Intelligent Robotic Systems	Intelligent Sensing Systems	Practical Language Processing	Intelligent Financial Risk Management	Architecting AI Systems
Reasoning Systems 5 Days	Intelligent Sensing and Sense Making 4 Days	Robotic Systems 5 Days	Vision Systems 5 Days	Text Processing using Machine Learning 4 Days	Advanced Machine Learning for Financial Services 4 Days	Explainable and Responsible Artificial Intelligence 3 Days
Cognitive Systems 3 Days	Pattern Recognition and Machine Learning Systems 5 Days	Autonomous Robots and Vehicles 5 Days	Spatial Reasoning from Sensor Data 3 Days	New Media and Sentiment Mining 3 Days	Explainable & Responsible AI for Finance 3 Days	AI and Cybersecurity 3 Days
Machine Reasoning 4 Days	Problem Solving using Pattern Recognition 5 Days	Human-Robot System Engineering 4 Days	Real Time Audio-Visual Sensing and Sense Making 4 Days	Conversational UIs 4 Days	Credit Risk Modelling and Analytics 4 Days	Architecting Agentic AI Solutions 4 Days
				Text Analytics 3 Days	Alternative Data for Fintech Innovation 3 Days	Integrating and Deploying AI Solutions 3 Days
Practice Module (10 man days)	Practice Module (10 man days)	Practice Module (10 man days)	Practice Module (10 man days)	Practice Module (10 man days)	Practice Module (10 man days)	Practice Module (15 man days)
Graduate Certificate in Intelligent Reasoning Systems	Graduate Certificate in Pattern Recognition Systems	Graduate Certificate in Intelligent Robotic Systems	Graduate Certificate in Intelligent Sensing Systems	Graduate Certificate in Practical Language Processing	Graduate Certificate in Intelligent Financial Risk Management	Graduate Certificate in Architecting AI Systems

Graduate Certificate in Pattern Recognition systems

This graduate certificate teaches how to design and build systems that make decisions by recognising complex patterns in data. Examples are robotic systems and smart city applications that take as input diverse sensor data streams. These systems will utilise the latest pattern recognition, machine learning and sensor signal processing techniques.

Key Takeaways:

- Build intelligent pattern recognition systems
- Understand core and advanced pattern recognition techniques and gain experience applying these techniques in practical systems and applications
- Understand and apply advanced signal processing techniques to sensor data
- Be familiar with current best practices and tools for building pattern recognition systems

Practice Module : Objectives

The objective of this practice module is threefold:

- Firstly, to expose participants in a supervised manner to real world problems so that they may practice the use of the skills they have learned during the individual course modules in a real world setting and obtain expert advice and guidance when needed.
- Secondly, to enable participants to demonstrate their proficiency across all of the skills that they have learned in the course modules and hence be certified as competent at the Certificate level.
- Thirdly, to provide a formal grading mechanism so that the certificate may be used as one component in the NUS-ISS Stackable Master of Technology (MTech) in Artificial Intelligence Systems.

Graduate Cert: Assessment Components

- The graduate cert assessment comprises the assessment components below:

Assessment Component	Weight
Practice Module Project work (documentations and MVP* deliverables)	50%
Examination	50%

* MVP: minimal viable product of pattern recognition system

Standard Grading Scheme		CAP
A+	85 - 100	5.0
A	80 - 84	5.0
A-	75 - 79	4.5
B+	70 - 74	4.0
B	65 - 69	3.5
B-	60 - 64	3.0
C+	55 - 59	2.5
C	50 - 54	2.0
D+	45 - 49	1.5
D	40 - 44	1.0
F (Fail)	0 - 39	0.0

- A participant must attain a minimum overall score of **50%** in order to pass the practice module and hence be awarded the Graduate Certificate in Pattern Recognition Systems.

Project Work: Assessment

Project Work Assessment Component	Weight
First Presentation	5%
Final Presentation	10%
Final Report	15%
Final System	15%
Peer Review	5%
Total	50%

Practice Module : Timeline

The practice module will take an estimated 10 days of effort by participants. These days are not expected to be continuous and may stretch over many weeks. The overall agenda is shown below.

Day 1: Introduction, project initialization

Day 2 to 9: First presentation, project work, supervision and feedback

Day 10: Project final video presentation and report submission

Written examination (3 hours open book)

Practice Module : Timeline : Activities

Day	Activities	Remarks
Day 1	Introduction to the practice module Participants will be made fully aware of the practice module requirements and assessment process.	
Days 2 - 9	Project work, supervision and feedback - Participants will work on their projects independently. Participants will be able to meet with ISS course lecturers to obtain advice and guidance. First Presentation - Participants will make a presentation in which they outline the goals of their projects along with details of the data resources required/available, techniques/tools used, progress, etc.	Submission of Project Proposal (Deadline: 15 Sep 2026) First Presentation (Date: 30 Sep 2026, 6.30-10.30pm, via zoom)
Day 10	Project final video presentation - Participants will submit a final project video presentation in which they describe fully the project they have undertaken and the methods and metrics they have used to evaluate its success. Project Work Assessment - Participants will submit a final project report including other relevant project deliverables, e.g. runnable software system.	Submission of deliverables (Deadline: 31 Oct 2026)
Exam Day	Written Examination Participants will undertake a formal written examination with a scope covering all of the topics taught in the course modules.	Examination

Practice Module : Project Work

Requirements:

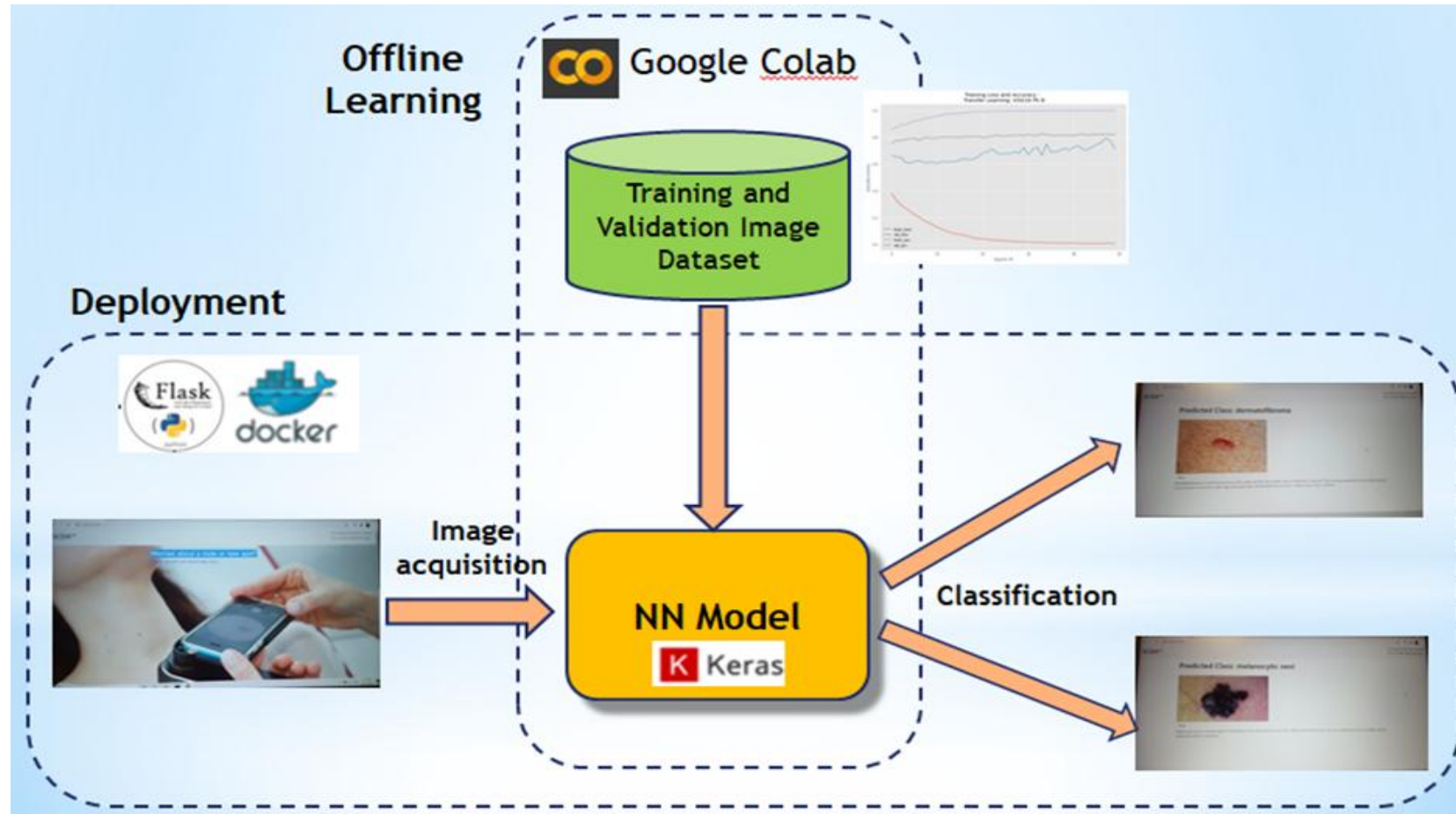
- Form a project team of max 5 members and enrol in Canvas project groups.
- The team may work on any practical application that demonstrates the advantage of pattern recognition and machine learning techniques.
- A suitable project uses AI/machine learning techniques to design and build a pattern recognition system to solve a real-world problem using the skills taught in the 3 course modules.
- The project must develop, integrate and demonstrate at least **three** out of following aspects:
 - ❖ Supervised learning / unsupervised learning scenarios
 - ❖ Machine learning/ Deep learning techniques
 - ❖ Hybrid machine learning /Ensemble approach
 - ❖ Intelligent sensing / sense making techniques

Practice Module : Project Work

Project Examples:

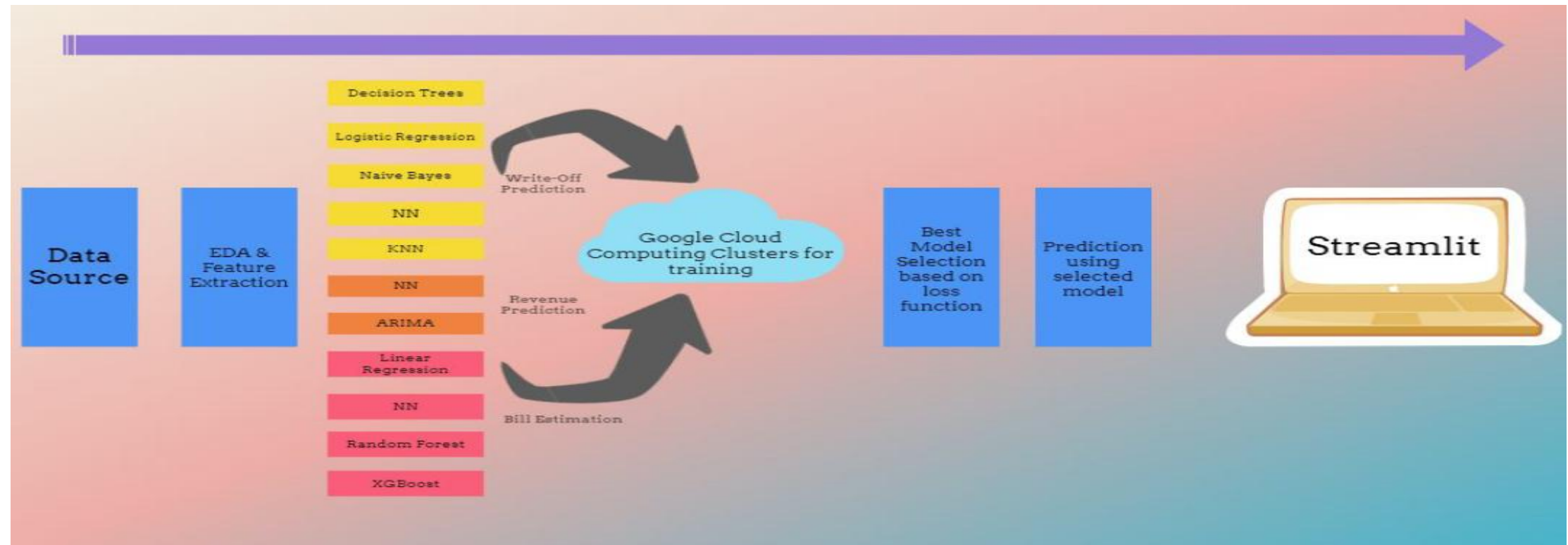
- Stock market forecasting
- Speech to text transcriber
- Facial emotion recognition
- Automatic image caption generation
- Hand gesture recognition
- Mini social robots
- Skin cancer detection
- Defect detection system

Project Example: Skin Cancer Detection



Project Example: Private Healthcare Administration Use Cases

- Use Case #1: Prediction of Write-Off Cases
- Use Case #2: Prediction of Inpatient Revenue
- Use Case #3: Hospital Bill Estimation Upon Admission



Project Example: Crowdedness Prediction

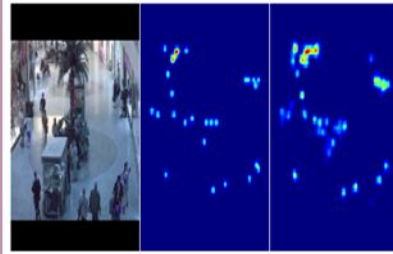
No. of ppl: 23



Image Regression

- Directly predict number of persons based on image
- 2 Models built using Keras Tensorflow
 - Sequential CNN
 - ResNet

Label = 30.00 Pred = 27.20



Density-Based Approach

- Models developed generate density maps
- U-Net Model architecture
- Different depths explored
 - 4 – encoder
 - 3 – encoder



Object Detection Approach

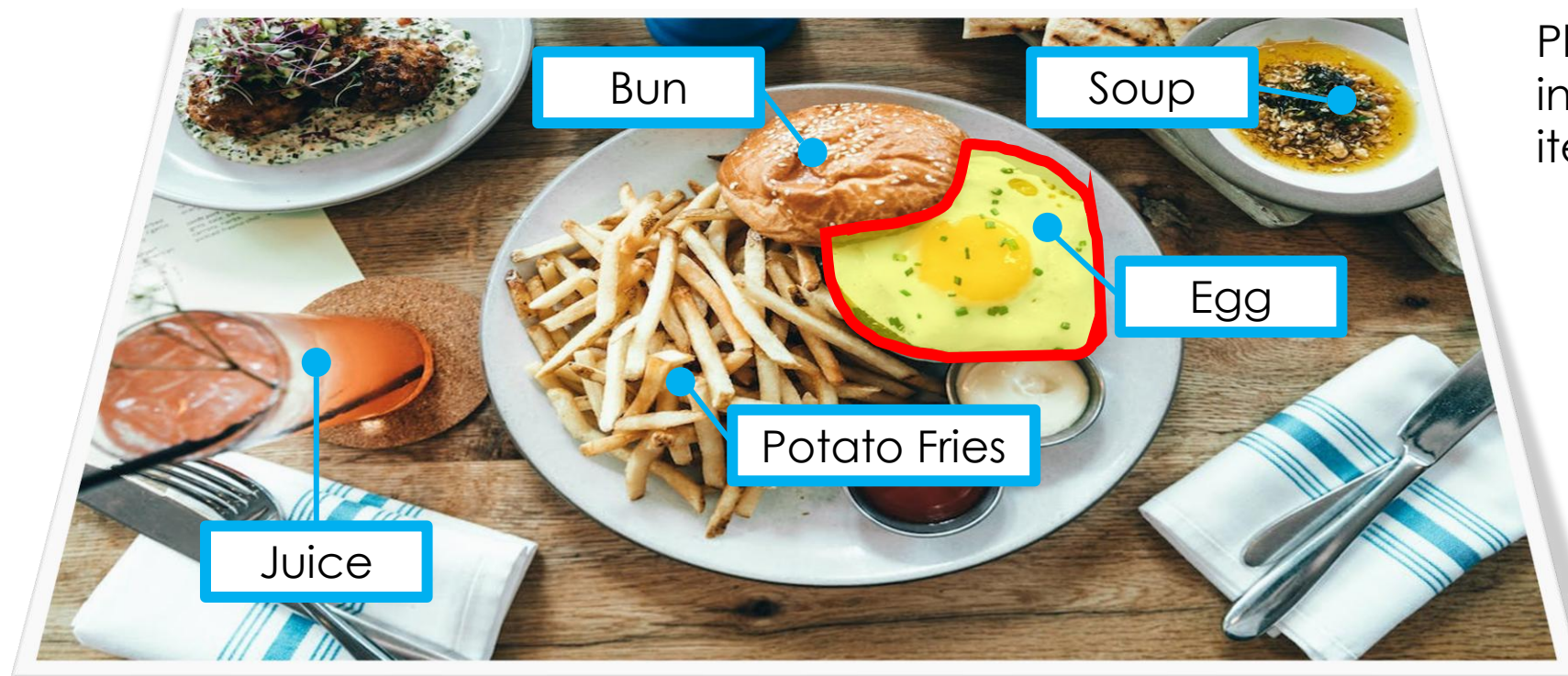
- Leveraged on pre-trained model architecture – yolov5s
- Trained using custom data, i.e. bounding box that encapsulates person's head



Image Classification

- Trained models that classify into three classes of crowdedness levels
 - Sparse
 - Medium
 - Crowded
- 2 Models built
 - Sequential CNN
 - ResNet

Project Example: FoodieVision



Phase 1 – Identify individual food items

Phase 2 – Identify food segmentation

Phase 3 – Natural question/answer interface

Practice Module: Project Work

Project Deliverables:

- A runnable pattern recognition system
- Datasets
- Final report to describe:
 - Tools/techniques you have used
 - System design / Models
 - System performance
 - Findings and discussions
- Python/R/Java/... codes, model files, other supporting documents (if any)
- A video presentation file, .mp4/.mov/.wmv etc., containing a 10 -15 mins presentation,
- **Slides of two presentations**

Practice Module: Project Work

Additional Submissions:

- 1-2 pages individual project report per project member, including:
 - Individual reflection of project journey: (1) personal **contribution** to group project (2) what learnt is most **useful for you** (3) how you can apply the knowledge and skills in **other situations or your workplaces**
- Submit Peer Review through the peer review system.

Submission deadline: 31/10/2026

Please submit to Canvas >>> Assignments >>> Practice Module

Please submit only one ZIP file from each team except for the individual report.

Questions & Answers



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